

FINAL INTERNSHIP PROGRESS REPORT

Week 7: Production-Grade SOC Dashboard, Thread-Safe IPC & Secure Active Response Engineering

Lê Quý Tôn - Student ID: 23BI14425

June 6, 2026

Project: CyberSec Internship - Cygnus-Security
Specialty: Cyber Security (USTH Academic Year 2023–2026)
Related Issues: #30 & #31 (Final Milestone)

1. Contribution Scope

This final report details the comprehensive optimization, hardening, and graphical visualization of the closed-loop Intrusion Detection and Prevention System (IDPS) developed over the internship period. Work in Week 7 shifts the architecture into a production-ready deployment by engineering a real-time Security Operations Center (SOC) monitoring dashboard via Streamlit and Altair. Crucially, the backend pipeline (`test.py`) was hardened using mutual exclusion primitives (`threading.Lock`) to prevent concurrent race conditions during telemetry synchronization, and legacy command execution gates were refactored to eradicate Command Injection vulnerabilities.

2. Objectives of This Week

- Develop an asynchronous telemetry data exporter (`export_dashboard_telemetry`) to securely pipe live network state spaces into a standardized JSON data buffer.
- Implement a robust mutual exclusion primitive layer (`state_lock`) to eliminate memory-corruption hazards and race conditions during dictionary size mutations.
- Harden the active response mitigation gateway by replacing vulnerable shell invocation APIs with decoupled parameter arrays via `subprocess.run`.
- Optimize the geometric layout of the SOC dashboard, forcing a clean horizontal 0-degree rendering alignment for host IP text labels on the X-axis under Altair Engine.

3. Detailed Work & System Architecture

3.1 Thread-Safe Inter-Process Communication (IPC)

To bridge the asynchronous gap between the non-blocking log parsing loop and the Streamlit UI without causing thread collisions, a lightweight telemetry data store (`dns_telemetry.json`) was deployed.

To eliminate the critical `RuntimeError` caused by concurrent read-write access to global monitoring structures (such as when the asynchronous unblocking timer mutates data while the exporter loop scans it), a mutual exclusion lock (`state_lock`) was integrated. The telemetry serialization process was refactored as follows:

```
def export_dashboard_telemetry():
    telemetry_file = "dns_telemetry.json"
    with state_lock:
        active_blocks = list(blocked_ips)
        serialized_quota = {
            f"{ip} -> {domain}": bytes_count
            for (ip, domain), bytes_count in volume_quota.items()
            if bytes_count > 0
        }
        data_to_dump = {
            "blocked_ips": active_blocks,
            "volume_quota": serialized_quota,
            "last_updated": time.strftime("%Y-%m-%d %H:%M:%S")
        }
        with open(telemetry_file, "w") as f:
            json.dump(data_to_dump, f, indent=4)
```

3.2 Active IPS Hardening: Secure Execution Boundary

To align the framework with secure application design patterns, all legacy invocations using `os.system()` were completely excised. Passing dynamically parsed network strings (such as fields extracted from `dns.rrname`) directly to a shell interpreter exposes the gateway infrastructure to dangerous OS Command Injection attacks.

The enforcement tier was refactored to execute binary parameters inside decoupled arrays, guaranteeing strict execution boundaries at the kernel level:

```
# Secure architectural deployment of Linux Netfilter rules
subprocess.run(["sudo", "iptables", "-I", "FORWARD", "1", "-s",
    ip_address, "-j", "DROP"], check=False)
subprocess.run(["sudo", "iptables", "-I", "INPUT", "1", "-s",
    ip_address, "-j", "DROP"], check=False)
```

3.3 Frontend Layout & X-Axis Geometric Correction

The Streamlit interface (`dashboard.py`) was customized to operate efficiently over a localized network topology socket. Standard chart widgets auto-rotate X-axis text values to 90 degrees when encountering string data vectors, rendering host IP labels difficult to interpret.

This layout flaw was resolved by shifting the rendering pipeline from native charts to the advanced Altair graphical engine, explicitly binding the `labelAngle` axis constraint property to 0 degrees and constraining bar chart width profiles to 40 pixels for sharp, concise telemetry tracking:

```
chart = alt.Chart(quota_df).mark_bar(color="#1f77b4", size=40).
    encode(
        x=alt.X("Source Host IP:N", axis=alt.Axis(labelAngle=0, title
            ="Source Host IP Address")),
        y=alt.Y("Volume (Bytes):Q", title="Volume (Bytes)")
    ).properties(height=400)
st.altair_chart(chart, width='stretch')
```

4. Comprehensive Evaluation & Results

Integrated stress testing was executed utilizing an adaptive data exfiltration script (`test.sh`) streaming 10-byte micro-sliced hexadecimal chunks with a 3-second delay, designed to bypass machine learning lexical controls.

- **Evasion Escalation Tracking:** For the initial 990 bytes of accumulated transmission, the machine learning tree registered safe states due to short payload lengths ($S_{\text{Lexical}} = 0$). However, the dynamic behavioral correlator continuously monitored structural data volumes alongside temporal sliding window footprints.
- **Automated Mitigations Commitment:** Exactly as the accumulated volume broke the absolute protection boundary ($Volume \geq 1000B$) and the persistent beaconing score was validated ($Temporal \geq 20$), the anti-evasion override instantly force-escalated the unified risk profile score to a critical ceiling value of **99/100**.
- **SOC Telemetry Real-Time Update:** Within 0.2 seconds of the escalation, the Streamlit dashboard rendering under the Network URL updated its presentation states smoothly: the **Isolated Hosts** metric card incremented to 1, the bar graphs cleanly mapped the breached IP usage footprint, and the kernel-level rule table populated critical network dropping states. Simultaneously, the execution console of the attacker script logged an immediate socket freeze with **[BLOCKED]** notifications.

5. Methodological Analysis (Defensive Value)

The successful integration of the dynamic multi-indicator risk matrix proves that stateful correlation is superior to static threshold gates. Advanced persistent threat (APT) actors routinely bypass static security boundaries by implementing time-dilation and micro-slicing tactics.

By mapping disjointed low-entropy indicators into a unified scoring pool ($R = S_{\text{Lexical}} + S_{\text{Temporal}} + S_{\text{Volumetric}}$), the system forces independent stealth vectors to compound. Legitimate, high-volume transactions (such as software updates over DNS) register high volume but zero temporal mutations, thereby safely clearing the system without triggering false positives. Conversely, anomalous data channels are captured with high precision, transforming the gateway from a basic stateful filter into an adaptive behavioral profiling node.

6. Final Synthesis of Issues & Fixes

- **Issue:** Reading dictionaries during asynchronous unblocking timers threw concurrent modification runtime errors.
- **Fix:** Wrapped all dictionary traversal, extraction, and update sequences across both files within a thread-safe `state_lock` context manager block.
- **Issue:** Standard string labels on the dashboard graph axis auto-rotated vertically, causing severe layout distortion on wide monitors.
- **Fix:** Migrated the rendering pipeline to the Altair engine, overriding axis label properties to a fixed 0-degree horizontal vector and setting manual column widths.

7. Conclusion & Thesis Readiness Statement

The core operational milestones of the internship project have been fully achieved. The engineered IDPS successfully demonstrates protocol decoding, L7 deep packet inspection, hybrid machine learning heuristic routing, stateful behavioral tracking, and secure kernel-level automated enforcement. The data models, code execution artifacts, and telemetry graphs gathered during this final phase provide complete, mathematically validated datasets. The technical foundation is verified as completely robust and structurally ready for transition into the final Bachelor's Thesis drafting phase at USTH.